MILITARY SPECIFICATION

TOLNAFTATE POWDER

This specification is approved for use by the Defense Personnel Support Center, Defense Logistics Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Tolnafate Powder (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a powder containing Tolnaftate, USP, 1 percent, Talc, USP, 10 percent, and Corn Starch, 89 percent.

3.1.1 Assay for tolnaftate. Shall contain not less than 90.0 percent and not more than 115.0 percent of the required amount when determined as specified in 4.7.1.

3.1.2 Identification of tolnaftate. The Rf value of the main spot obtained from the test solution corresponds to that obtained from the USP reference standard solution when determined as specified in 4.7.2.

3.2 Ingredients.

3.2.1 Tolnaftate. The tolnaftate used in the manufacture of the finished product shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.2.2 Talc. The talc used in the manufacture of the finished product shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.2.3 Corn starch. The corn starch used in the manufacture of the finished product shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto, except that the pH shall be not less than 4.0 and not more than 7.0, when determined by the USP method.

3.3 Expiration dating period. The expiration dating period shall be not less than 36 months.

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3.3.1 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

3.4 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of finished powder produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Units of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

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4.3.1 End item testing. Noncompliance with any of the following is cause for rejection. The acceptance number for each test shall be zero (0).

TABLE II. Criteria for end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Assay for tolnaftate	3.1.1	S-2	5 Composite
Identification of tolnaftate	3.1.2	S-2	5 Composite

1/ Where possible, the sample shall be used for two or more tests.

4.4 Examination. The finished powder shall be examined to determine compliance with all requirements of this specification. Non-conformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Powder not essentially free from foreign matter.
102	Powder not grit-free.
103	Powder not homogeneous.
104	Sifter or dispenser top not functioning properly.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks or tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests and examinations. Tests and examinations shall be performed to determine compliance with all requirements of this document and all referenced documents.

4.7.1 Assay for tolnaftate.

Weigh accurately about 50 mg of tolnaftate reference standard, and transfer quantitatively into a 100 ml volumetric flask. Dissolve in and dilute to volume with methanol. Pipet 4.0 ml into a 200 ml volumetric flask, and dilute to volume with methanol.

Note: Retain remainder of first dilution for Identification test.

Weigh accurately about 1.00 gram of sample, into a 125 ml conical flask, add 70 ml of methanol, and heat the mixture on a steam bath for 10 minutes.

Cool to room temperature, filter with vacuum through a medium-porosity, sintered-glass funnel into a 100 ml volumetric flask (use a Fisher Filtrator or equivalent); rinse the funnel and residue with small portions of methanol.

Add the rinse filtrates to the flask, dilute to volume with methanol. Pipet 10.0 ml into a second, 100 ml volumetric flask, and dilute to volume with methanol.

Note: Retain remainder of former dilution for Identification test.

Obtain the absorption spectrum (from 240 nm to 280 nm) of sample and standard solutions with a spectrophotometer zeroed to methanol in a 1 cm cell. Record the value of maximum absorbance (at about 260 nm) of sample and standard.

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Calculation:

$$\frac{Au}{As}\left(\frac{\text{weight standard (mg)}}{100} \times \frac{4}{200}\right)\left(\frac{100}{\text{weight sample (grams)}} \times \frac{100}{10}\right) = \text{mg tolnaftate per gram}$$

where:

Au = Absorbance sample.
As = Absorbance standard.

4.7.2 Identification.

Evaporate 2.0 ml of standard dilution and 10.0 ml of sample solution in separate, labeled vessels: Standard and Sample. Dissolve each residue in 1 ml of benzene-alcohol mixture (1:1) just prior to use.

Using a suitable thin layer chromatography apparatus proceed as follows:

Adsorbant	Silica Gel G
Solvent (Mobile Phase)	Toluene
Application of Sample and Standard	Apply 10 µl of each
Migration Period	Approximately 35 minutes
Detection of Spots	Use short wave ultraviolet light source.
Spray	Use 1 percent iodine solution (1 gram in 100 ml carbon tetrachloride). Permit to dry.
Observation	Use visible light source.

Compare migration rates of standard and sample.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 and as specified herein.

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5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One box containing six 4 oz bottles, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available immediate container with sifter or dispenser top considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. Each unit shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Six (6) units of issue shall be packaged as specified in 5.2.9 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Six (6) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Six (6) units of issue shall be packed as specified in 5.4.3.3 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

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5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number need not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type II shelf life markings as specified in MIL-STD-129 shall be shown.

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type II shelf life marking as specified in MIL-STD-129 shall be shown.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-01-037-6536	TOLNAFTATE POWDER, 1%, 45 gram, 6s

Preparing activity:
DLA-DM

Project No. 6505-1990

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DOCUMENT IDENTIFIER (Number, AND TITLE

MIL-T-37844(DLA-DM TOLNAFTATE POWDER

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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